Material cases: Exercises

> Reference: Jennifer Rupp, Daniele Ielmini, Ilia Valov (eds.), Resistive Switching: Oxide Materials, Mechanisms, Devices and Operations, Springer, 2022. DOI: 10.1007/978-3-030-42424-4.

Unless stated otherwise, all numerical values below describe hypothetical models, not measured device specifications.

E01 Diffusion scale

In an ideal one-dimensional unbiased point-source diffusion model, $D = 10^{-21}\,\mathrm{cm^2/s}$. How long until the rms displacement is $20\,\mathrm{nm}$? Can this be used as a measured SET time?

E02 Parallel RC

For a linear, small-signal, unchanged-state model with $R = 1\,\mathrm{M\Omega}$ in parallel with $C = 10\,\mathrm{pF}$, calculate $\tau = RC$ and the frequency of the maximum of $-\operatorname{Im} Z$. Does fitting this model uniquely determine a filament morphology?

E03 Weibull probability and median

Assume $X = |V_{\mathrm{RESET}}|$ is Weibull with $\eta = 1\,\mathrm{V}$, $\beta = 4$. Calculate $F(0.8\,\mathrm{V})$ and the median.

E04 Weibull quantiles and data

For the E03 distribution calculate the 10% and 90% quantiles. Explain why 1,000 successive cycles of one device are not automatically 1,000 independent devices; name another important reporting condition.

E05 Heating and scaling

Assume steady $T = T_0 + R_{\mathrm{th}} V^2/R$, with $R = 150\,\Omega$, $V = 0.9\,\mathrm{V}$ and $R_{\mathrm{th}} = 3 \times 10^4\,\mathrm{K/W}$. Find power and temperature rise. Under what assumptions can the RESET voltage be independent of R?

E06 HfO_2/Ti aging evidence

A specified group's HRS/LRS ratio falls from 10^4 to 10^2. By what factor did it decrease? Can this alone determine the changes of HRS and LRS individually? If separate fresh devices from the same batch are measured at different times, and day 1 and day 60 depth-profile XPS measurements are compared, what interpretation can this support? Give at least two limitations.

E07 Classical nucleation

Assume $\Delta G = f(4\pi r^2\gamma - 4\pi r^3|\Delta g_v|/3)$, $\gamma = 0.03\,\mathrm{J/m^2}$, $|\Delta g_v| = 3 \times 10^7\,\mathrm{J/m^3}$ and $f = 0.1$. Find r^* and the barrier in joules and eV using $1\,\mathrm{eV} = 1.602 \times 10^{-19}\,\mathrm{J}$.

E08 Thermal factor and rate

Use E07's barrier and $k_B T = 0.02585\,\mathrm{eV}$ at 300 K to estimate $e^{-\Delta G^*/k_B T}$. Why is this not itself a rate in s^{-1}? Give two limitations of the classical model.

Material cases: Exercises (continued)

E09 Fast pulse

Suppose $C = 0.2\,\mathrm{pF}$ and the applied voltage increases at $1\ \mathrm{V/ns}$. At $0.5\,\mathrm{ns}$ no conductive bridge has formed; at $1.2\,\mathrm{ns}$ a distinct conduction-current rise begins. Compute the ideal capacitive current before conduction and the SET time using pulse start and this rise as the specified endpoints. Why can the current peak alone not determine device energy?

E10 Three-terminal switch

Compare the conductive path of a two-terminal cell with the roles of gate, source and drain in a three-terminal atomic-switch example. Why avoid a conductive path linking the gate to source or drain? Why need this not operate like an ordinary MOSFET?

E11 CBRAM reactions and humidity

Write illustrative oxidation and reduction half-reactions for a Cu-active cell. Can positive Cu-electrode bias alone predict where the metallic bridge grows? In a specific Cu/SiO_2/Pt study, suppose humidity changes ion concentration while estimated diffusivity remains approximately unchanged. Why does this not establish that humidity always increases ionic mobility?

E12 Evidence and alternatives

A device exhibits similar I–V hysteresis at two temperatures. Design at least two controlled comparisons to distinguish metal nucleation/transport, thermally assisted processes and transient parasitic effects. State what each result could support and what it cannot establish on its own.

E13 Conditional field drift

Assume a dilute singly charged carrier, applicable linear response and Einstein relation $\mu = D/(k_B T/e)$, $D = 10^{-21}\,\mathrm{cm^2/s}$, $k_B T/e = 0.02585\,\mathrm{V}$, and constant field $E = 10^5\,\mathrm{V/cm}$. Calculate mobility, drift speed and the distance/speed scale for 20 nm. Why do the assumed relations need separate justification, and why is the latter not a measured SET time?

E14 RESET current bins

For six hypothetical Ohmic bins with $R_{\mathrm{ON}} = 100, 150, 200, 250, 300, 400\,\Omega$, assume the peak voltage magnitude is fixed at $0.9\,\mathrm{V}$ and $I_{\mathrm{peak}} = |V|/R_{\mathrm{ON}}$. These are not a reconstruction of the published Weibull distribution. Calculate the six currents. Under a lumped thermal model, which conditions would allow the same critical temperature to produce an approximately constant peak voltage?

E15 Ti-thickness comparison

For the specified HfO_2/Ti comparison, nominal Ti thicknesses are 10, 30, and 100 nm; their roughly 60-day HRS/LRS ratios are 5×10^4, 10^2, and 10. Compute the ratio between the first and last. Why must device selection, read convention and different layer structures be kept separate before attributing the difference to Ti thickness?

Material cases: Exercises (continued)

E16 Competing timing laws

Suppose both a simplified nucleation model and a high-field migration model give $\ln(t/\mathrm{ns}) = 6 - 2(V/1\,\mathrm{V})$ in the interval 1–2 V. The nucleation barrier is $\Delta G(V) = \Delta G_0 - 0.05\,\mathrm{eV}\,(V/1\,\mathrm{V})$ with $\Delta G_0 > 0.10\,\mathrm{eV}$. Calculate the times at 1, 1.5 and 2 V. If migration has $v = v_0 \exp[eaV/(2k_BTL)]$ and $t = L/v$, verify the slope using $a = 0.3\,\mathrm{nm}$, $L = 3\,\mathrm{nm}$ and $k_BT/e = 0.025\,\mathrm{V}$. Does equal slope identify the mechanism?

E17 Hypothetical multilevel programming

Assume $R_{\mathrm{ON}} = A/I_{\mathrm{prog}}$ with $A = 0.5\,\mathrm{V}$, and SET compliance currents of 5, 10 and 20 μA. Calculate the nominal resistances and their read currents at 0.1 V assuming Ohmic, non-disturbing reads. Why do three nominal values alone not establish a reliable multilevel memory?

E18 Radiation and evidence

In a reported Cu/SiO_2 study, ten $10 \times 10\,\mu\mathrm{m}^2$ devices were cycled ten times at different ^{60}Co doses, with a maximum of $7.1\,\mathrm{Mrad(SiO_2)}$. Three devices were also programmed over multiple compliance currents before and after exposure. If plotted HRS/LRS distributions and programming curves are similar, what does this support? Explain two limits on interpreting the result as a reliability guarantee.

E19 Ta_2O_5 film density and humidity

For a reported film-density comparison, a $7.6\,\mathrm{g/cm^3}$ film retained nonvolatile switching across the tested RH variation, while a $7.1\,\mathrm{g/cm^3}$ film changed from nonvolatile below 50% RH to volatile above 50% and recovered on reducing RH; forming preceded each RH measurement, and example I–V conditions include 22% and 85% RH. Propose a reversible humidity control and explain why this does not prove proton conduction or a universal RH threshold. A separate Cu/Ta_2O_5/Pt cyclic-voltammetry study varied RH over 5–85% and plotted a $100\,\mathrm{mV/s}$ sweep: why should its fitted transport parameters not be silently attributed to the density-comparison devices?

E20 Spectroscopic and impedance evidence

During Ag/SiO_2/Au switching, plasmon-enhanced scattered spectra are consistent with residual particles and changing growth locations. Separately, a programmed $500 \times 500\,\mu\mathrm{m}^2$ Cu/SiO_2 device fits a parallel-RC model with approximately unchanged fitted capacitance over $230\,\Omega$–$55\,\mathrm{k}\Omega$. Explain the inference supported by each observation and why neither alone proves a unique microscopic filament geometry. Suggest an independent control or complementary measurement.

E21 Synaptic pulse readout

Assume an ideal synchronized series circuit with $V_1 = 1\,\mathrm{V}$, $R_s = 10\,\mathrm{k}\Omega$, and V_2 measured across R_s. Three successive samples have $V_2 = 0.20$, 0.25 and $0.30\,\mathrm{V}$; the device is approximately resistive at each sample and parasitics are negligible. Derive and calculate R_{dev} and its conductance at each sample. Why do these assumed samples not reproduce or prove the reported ten-pulse experiment, and why is the voltage during a programming pulse not automatically a nondisturbing read?